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Left Kan extensions and monotone-light factorizations

João Xarez

Department of Mathematics, University of Aveiro, Portugal

Let $Set^K \vdash Lan_K \colon Set^{\mathbb{B}} \to Set^{\mathbb{A}}$ be an adjunction given by left Kan extensions, and such that, for every object $A \in \mathbb{A}$, each comma category $(K \downarrow A)$ is a pseudo-filtered category, and all the morphisms in $\mathbb{B}$ are monomorphisms. The induced reflection from the adjunction above is "very-well-behaved", providing an associated monotone-light factorization system on $Set^{\mathbb{B}}$. The induced reflection $I \vdash H$ of $Set^{\mathbb{B}}$ into a full subcategory $Mono(Lan_K)$ is obtained by factorizing (componentwise) the unit morphisms of the given adjunction through surjections and injections. This new result just stated generalizes section 3 in [3]. In fact, if $\mathbb{B}$ is pseudo-filtered and $\mathbb{A} = \coprod_{i \in I} \mathbf{1}$ is the coproduct of a family of categories with just one object and one morphism (ordinal number one), it can be shown that this is a special case of the more general one above. In particular, for $\mathbb{A} = \mathbf{1}$ the left adjoint Lan_K becomes the colimit functor, and in the very simple case with $\mathbb{B} = \mathbf{2}$ (ordinal number two) the monotone-light factorization is non-trivial (that is, distinct from the reflective factorization system), as shown in section 3 in [3]; notice that, for instance, if $\mathbb{B} = \mathbf{2}$ and $\mathbb{A} = \mathbf{1} \sqcup \mathbf{1}$ then the monotone-light factorization is the same as when $\mathbb{A} = \mathbf{1}$.

Keywords: Monotone-light factorization; pseudo-filtered category; left Kan extensions; presheaves
Subject area: Category theory
MSC2020: 18A40, 18A32, 18E50

References

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[3] Xarez, J. J. *Very-well-behaved epireflections for categories of models of sketches*, Cahiers Top. Géom. Diff. Catég. LXVII(3), (2026) 355–387.